

Let $A = (a_{ij}) \in \mathbb{R}^{2 \times 2}$ be a positively definite matrix and $\langle, \rangle$ be a scalar product given by $\langle \vec{u}, \vec{v} \rangle = \vec{u}^T A \vec{v}$. Prove that there exists a basis of $\mathbb{R}^2$ such that it's orthogonal both with regard to the standard scalar product, and the scalar product given by A (from now A-product).

Let $B = \left(\begin{pmatrix} \cos \varphi \\ \sin \varphi \end{pmatrix}, \begin{pmatrix} -\sin \varphi \\ \cos \varphi \end{pmatrix} \right)$ then their standard scalar product is

$$\begin{pmatrix} \cos \varphi \\ \sin \varphi \end{pmatrix} \cdot \begin{pmatrix} -\sin \varphi \\ \cos \varphi \end{pmatrix} = (\cos \varphi, \sin \varphi) \begin{pmatrix} -\sin \varphi \\ \cos \varphi \end{pmatrix} = \cos \varphi \sin \varphi - \sin \varphi \cos \varphi = 0$$

However their A-product is

$$\begin{aligned} \left\langle \begin{pmatrix} \cos \varphi \\ \sin \varphi \end{pmatrix}, \begin{pmatrix} -\sin \varphi \\ \cos \varphi \end{pmatrix} \right\rangle &= (\cos \varphi, \sin \varphi) \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} -\sin \varphi \\ \cos \varphi \end{pmatrix} \\ &= -a \sin \varphi \cos \varphi + b \cos^2 \varphi - b \sin^2 \varphi + c \sin \varphi \cos \varphi \\ &= b(\cos^2 \varphi - \sin^2 \varphi) + \frac{c-a}{2} 2 \sin \varphi \cos \varphi \\ &= b \cos 2\varphi + \frac{c-a}{2} \sin 2\varphi \end{aligned}$$

If the basis is to be orthogonal with regard to the A-product, this expression must equal 0

$$\begin{aligned} 0 &= b \cos 2\varphi + \frac{c-a}{2} \sin 2\varphi \\ (a-c) \sin 2\varphi &= 2b \cos 2\varphi \\ \frac{\sin 2\varphi}{\cos 2\varphi} &= \frac{2b}{a-c} \\ \varphi &= \frac{1}{2} \arctan \left(\frac{2b}{a-c} \right) \end{aligned}$$

Assuming that $a \neq c$ $\arctan$ is defined for $\mathbb{R}$, so it will give us a solution for any A-product.

Assuming that $a = c$ simplifies the previous equation to

$$0 = b \cos 2\varphi$$

Which has an obvious solution of $\pi/4$.

Therefore the basis B is orthogonal both in regard to the A-product and the standard product if φ is chosen using the formulae given above.

Q. E. D.